

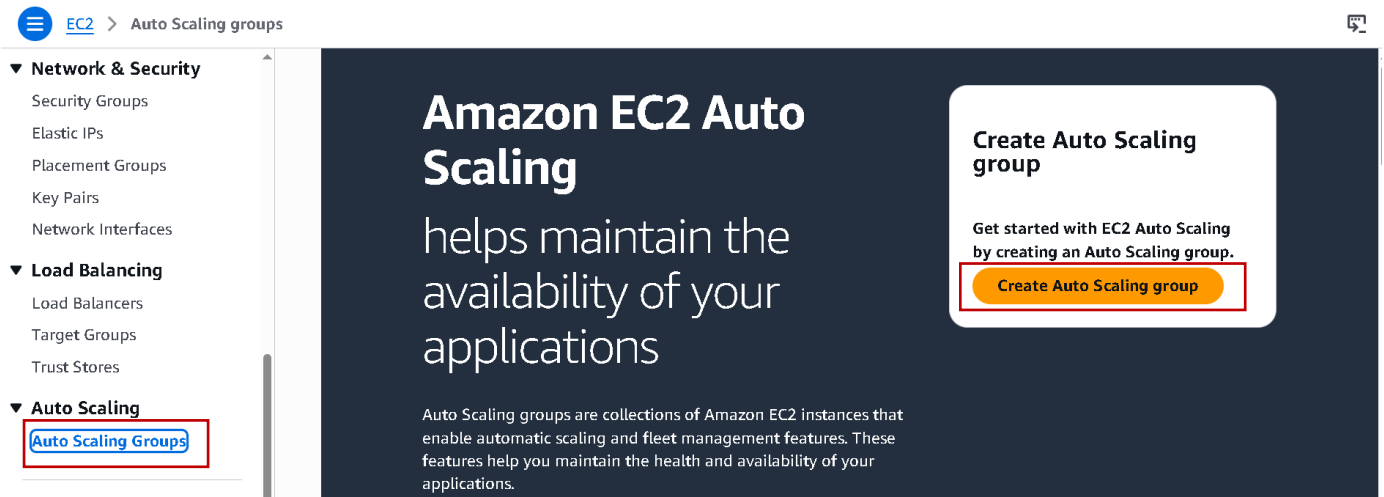
Manual Scaling: -

Summary: -

I worked on manual scaling by increasing or decreasing servers based on traffic requirements to maintain smooth application performance and availability.

Let's Start: -

- Now open your AWS console Search EC2 and open it.



- Now into Nav bar scroll down and click Auto scaling Groups.
- And then click Create Auto scaling group.

Choose launch template [Info](#)

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name

Auto Scaling group name

Enter a name to identify the group.

Must be unique to this account in the current Region and no more than 255 characters.

- Now enter Auto scaling group name.
- Scroll down.

Launch template [Info](#)

- i For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

[Create a launch template](#)

- Now we need to create launch Template.
- Understand launch templet and why we need to create with example – assume you have one web application running now during weekdays I have very normal traffic and I have only one EC2 instance which is handling all the traffic, and everything is going well.
- But during Saturday and Sunday you have faced issues because of lots of traffic coming on your web application.
- So, you decided to use Auto scaling and let's increases number of EC2 instances for Weekends.
- Now your requirement is during Saturday and Sunday your EC2 instance will be three and during weekdays only one EC2 instance.
- Now when we use Auto scaling that will add these EC2 instances for sure.
- but what will be the configuration of that EC2 (like I want t2 micro, extra-large instance, this type of operating system, this security group)
- And with launch template we can specify this type of configuration of EC2 on Weekends(Saturday-Sunday).
- Now click Create a launch template.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

LT-FOR-ASG

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '*', '@'.

Template version description

1

Max 255 chars

Auto Scaling guidance | [Info](#)

Select this if you intend to use this template with EC2 Auto Scaling

- ☐ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

- Enter launch template name.
- Enter version.
- And currently off the Auto scaling guidance.
- Scroll down

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose **Browse more AMIs**.

 Search our full catalog including 1000s of application and OS images

Recents

My AMIs

Quick Start

☒ Don't include in launch template

☐ Recently launched



[Browse more AMIs](#)

Including AMIs from
AWS, Marketplace and
the Community

- Now select your AMI click Browse more AMIs.

Choose an Amazon Machine Image (AMI)

[Cancel](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. You can select an AMI provided by AWS, our user community, or the AWS Marketplace; or you can select one of your own AMIs.

 Search for an AMI by entering a search term e.g. "Windows"



Quick Start AMIs (45)

Commonly used AMIs

My AMIs (6)

Created by me

AWS Marketplace AMIs (6082)

AWS & trusted third-party AMIs

Community AMIs (500)

Published by anyone

Refine results

[Clear all filters](#)

☐ Free tier only [Info](#)

▼ OS category

☐ All Linux/Unix

☐ All Windows

All products (45 filtered, 45 unfiltered)

< 1 >



Amazon Linux

Free tier eligible

Verified provider

Amazon Linux 2023 kernel-6.1 AMI

ami-009be0edec0817ffd (64-bit (x86), uefi-preferred) / ami-080cc69ee28841635 (64-bit (Arm), uefi)

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Platform: amazon

Root device type: ebs

Virtualization: hvm

ENA enabled: Yes

[Select](#)

☒ 64-bit (x86), uefi-preferred

☐ 64-bit (Arm), uefi

- Now select AMI I am selecting Amazon Linux.
- Simply click Select.

▼ Instance type [Info](#) | [Get advice](#)

[Advanced](#)

Instance type

t2.micro

Family: t2 1 vCPU 1 GiB Memory Current generation: true

On-Demand Windows base pricing: 0.017 USD per Hour

On-Demand RHEL base pricing: 0.0268 USD per Hour

On-Demand Linux base pricing: 0.0124 USD per Hour

On-Demand Ubuntu Pro base pricing: 0.0142 USD per Hour

On-Demand SUSE base pricing: 0.0124 USD per Hour

☐ All generations

[Compare instance types](#)

[Additional costs apply for AMIs with pre-installed software](#)

- Instance type t2 micro.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

demo

 [Create new key pair](#)

- Now choose Key pair.
- I have a key I use it if you not have simply create it.

▼ Network settings [Info](#)

Subnet [Info](#)

Don't include in launch template

 [Create new subnet](#) 

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone [Info](#)

Don't include in launch template

 [Enable additional zones](#) 

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Select existing security group

☒ Create security group

Security group name - *required*

web-server-2312

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255 characters.
Valid characters: a-z, A-Z, 0-9, spaces, and . _ : / 0 # , @ [] + = & ; ' ! \$ *

- Now select Create security group.
- And give name of group.

Description - *required* [Info](#)

Web Protect

VPC [Info](#)

vpc-072c4acd761c3b942
172.31.0.0/16

(default) ▼



- Enter description.

Inbound Security Group Rules

▼ Security group rule 1 (All, All, 0.0.0.0/0)

Remove

Type | Info

All traffic ▼

Protocol | Info

All

Port range | Info

All

Source type | Info

Custom ▼

Source | Info

🔍 Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - optional | Info

e.g. SSH for admin desktop

▼ Security group rule 2 (TCP, 22, 0.0.0.0/0)

Remove

Type | Info

ssh ▼

Protocol | Info

TCP

Port range | Info

22

Source type | Info

Custom ▼

Source | Info

🔍 Add CIDR, prefix list or security group

0.0.0.0/0 ✕

Description - optional | Info

e.g. SSH for admin desktop

- Now add two inbound security group.
- 1st type All traffic and sources Anywhere (0.0.0.0/0)
- 2nd type ssh and Source Anywhere (0.0.0.0/0)

▼ Advanced details Info

Domain join directory | Info

Select ▼

🔄 Create new directory ↗

IAM instance profile | Info

Select ▼

🔄 Create new IAM profile

Hostname type | Info

IP name ▼

DNS Hostname | Info

- So, for that scroll down and open Advanced details.
- And scroll down.

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash

yum update -y

yum install httpd -y

systemctl start httpd
systemctl enable httpd

echo "Welcome Ritesh Its Working"> /var/www/html/index.html
```

☐ User data has already been base64 encoded

Virtual server type (instance type)

t2.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage,

[Cancel](#)[Create launch template](#)

- Now here paste the script that provided in your resource.
- Now simply click Create launch template.
- Now its ready go back to your Auto scaling Group tab.

Launch template [Info](#)

Info For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template

Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.

LT-FOR-ASG

[Create a launch template](#)**Version**

Default (1)

[Create a launch template version](#)**Description**

1

Launch template

[LT-FOR-ASG](#)

lt-0e002022b82b2ff6e

Instance type

t2.micro

AMI ID

ami-009be0edec0817ffd

Security groups

-

Request Spot Instances

No

Key pair name

demo

Security group IDs

[sg-0a37415d3970732e4](#)

- Now simply click refresh and select created Launch template.
- Now scroll down and click next.

Network [Info](#)

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-072c4acd761c3b942
172.31.0.0/16 Default



[Create a VPC](#)

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets



aps1-az1 (ap-south-1a) | subnet-04ffa8e5e9c79ad3d
172.31.32.0/20 Default

aps1-az3 (ap-south-1b) | subnet-0bc545e3030c41ef4
172.31.0.0/20 Default

[Create a subnet](#)

- Now select Availability Zones and subnets because at launch template creation time we cant selected.
- So, select ap-south-1a and ap-south-1b
- Now scroll down and simply click next.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options



No load balancer

Traffic to your Auto Scaling group will not be fronted by a load balancer.



Attach to an existing load balancer

Choose from your existing load balancers.



Attach to a new load balancer

Quickly create a basic load balancer to attach to your Auto Scaling group.

- Now currently select No Load balancer.
- Right now we don't enable extra thing our main aim is implementing Manual Scaling.
- Simply scroll down click next.

Group size [Info](#)

Set the initial size of the Auto Scaling group. After creating the group, you can change its size to meet demand, either manually or by using automatic scaling.

Desired capacity type

Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)



Desired capacity

Specify your group size.

0

- Now Enter Desired Capacity 0.
- It means I am informing AWS right now I want 0 EC2 instance.

Scaling [Info](#)

You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits

Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

Equal or less than desired capacity

Max desired capacity

Equal or greater than desired capacity

Automatic scaling - *optional*

Choose whether to use a target tracking policy [Info](#)

You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.



No scaling policies

Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.



Target tracking scaling policy

Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

- Min desired capacity 0 and max desired capacity 5.
- Now scroll down and click next.

Add notifications - *optional* [Info](#)

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

[Add notification](#)[Cancel](#)[Skip to review](#)[Previous](#)[Next](#)

- Again, simply click next.

Add tags - *optional* [Info](#)

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

Tags (0)

No tags

[Add tag](#)

50 remaining

[Cancel](#)[Previous](#)[Next](#)

- Then again click next.

Step 6: Add tags

[Edit](#)

Tags (0)

Key	Value	Tag new instances
No tags		

[Preview code](#)[Cancel](#)[Previous](#)[Create Auto Scaling group](#)

- Now simply scroll down and click Create Auto scaling group.

Auto Scaling groups (1) [Info](#) Last updated 1 minute ago [Launch configurations](#) [Launch templates](#) [Actions](#) [Create Auto Scaling group](#)

Q Search your Auto Scaling groups

Name	Status	Launch template/configuration	Instances	Instance health	Desired capacity	Min	Max
ASG	At desired capacity	LT-FOR-ASG Version Default	0	-	0	0	5

- Now it's created if you go and check your instance currently no any EC2 instance is Created.
- Because our desired capacity is 0 and min capacity is 0.
- So now we update this.
- So simply open the ASG by clicking on it.

ASG

ASG Capacity overview [Edit](#)

arn:aws:autoscaling:ap-south-1:463646775279:autoScalingGroup:5b9dab94-099b-44c4-aece-f6018a7f90e1:autoScalingGroupName/ASG

Desired capacity 0	Scaling limits 0 - 5	Desired capacity type Units (number of instances)	Status At desired capacity
------------------------------	--------------------------------	---	--------------------------------------

Date created
Fri May 15 2026 15:13:12 GMT+0530 (India Standard Time)

- Now click on Edit.

Group size

Specify the size of the Auto Scaling group by changing the desired capacity. You can also specify minimum and maximum scaling limits.

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

1

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity

1

Equal or less than desired capacity

Max desired capacity

5

Equal or greater than desired capacity

[Cancel](#) [Update](#)

- Now Enter desired 1 and min 1 and click Update.

- Now go to instance and see one instance is created or not.

The screenshot shows the AWS Management Console 'Instances' page. At the top, there's a header with 'Instances (1/1)' and 'Info'. Below it, a search bar and a table of instances. The table has columns: Name, Instance ID, Instance state, Instance type, Status check, Alarm status, and Availability Zone. One instance is listed with ID 'i-088293f2aabffd264', state 'Running', type 't2.micro', and '2/2 checks passed'. Below the table, the 'Details' tab is selected for the chosen instance, showing its summary, public IPv4 address (65.0.4.135), and private IPv4 addresses (172.31.35.13).

- You see that one instance is created.
- Now click it and copy the public IPv4 address and paste in new tab.
- And check is working or not.



Welcome Ritesh its Working

- Now you see its working.

The screenshot shows the AWS Management Console 'Instances' page with the 'Instance state' dropdown menu open. The menu options are: Stop instance, Start instance, Reboot instance, Hibernate instance, and Terminate (delete) instance. The 'Terminate (delete) instance' option is highlighted. The instance table below shows the same instance as before, with state 'Running'.

- Now I got to instance tab and click instance state and click Terminate.
- Then this instance is deleted.
- But we set in our ASG min 1 so another 1 is automatically created.
- And if we want scale in and scale out we need to do it manually go inside Created Auto Scaling group and Edit capacity.